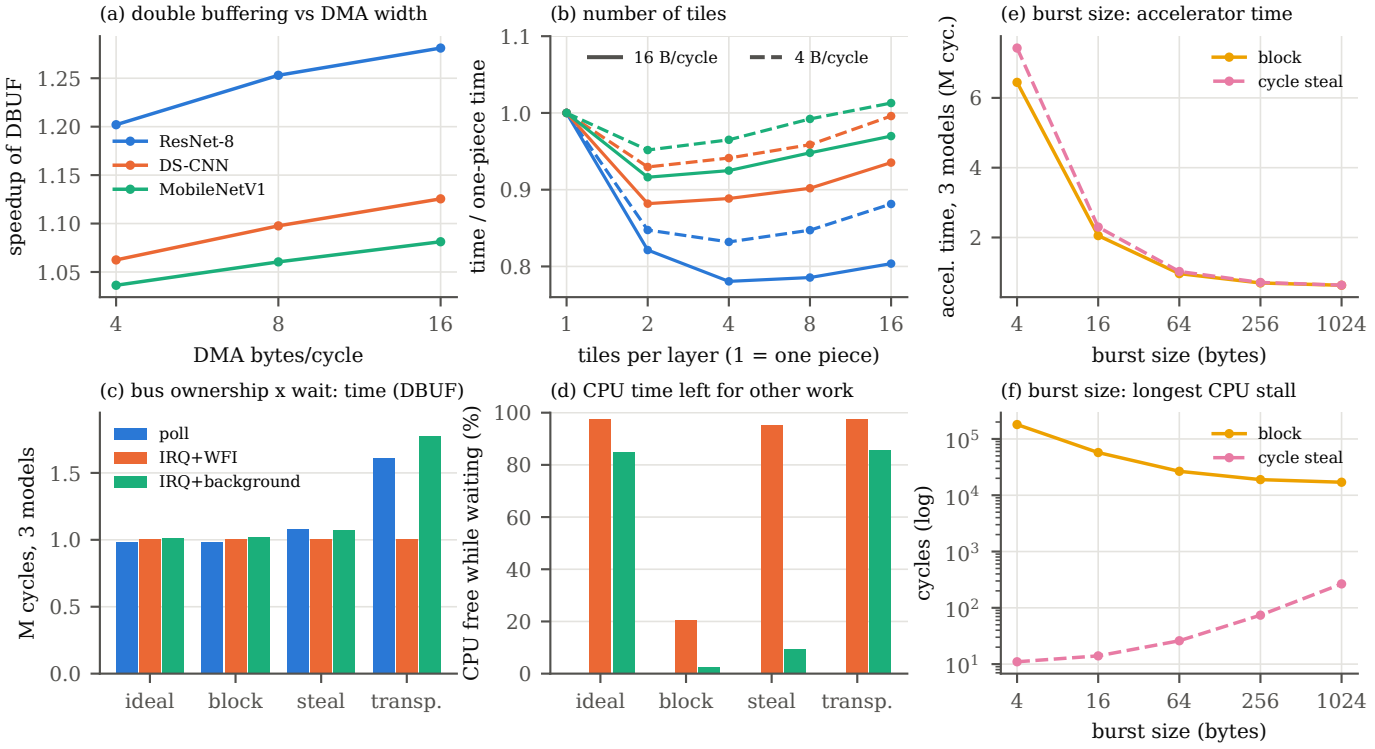
(a) double buffering vs DMA width
speedup of DBUF
1.25
1.20
1.15
1.10
1.05
ResNet-8
DS-CNN
MobileNetV1
4
8
16
DMA bytes/cycle
(b) number of tiles
time / one-piece time
1.1
1.0
0.9
0.8
16 B/cycle
4 B/cycle
1
2
4
8
16
tiles per layer (1 = one piece)
(e) burst size: accelerator time
accel. time, 3 models (M cyc.)
6
4
2
block
cycle steal
4
16
64
256
1024
burst size (bytes)
(c) bus ownership x wait: time (DBUF)
M cycles, 3 models
1.5
1.0
0.5
0.0
poll
IRQ+WFI
IRQ+background
ideal
block
steal
transp.
(d) CPU time left for other work
CPU free while waiting (%)
100
80
60
40
20
0
ideal
block
steal
transp.
(f) burst size: longest CPU stall
cycles (log)
10^5
10^4
10^3
10^2
10^1
block
cycle steal
4
16
64
256
1024
burst size (bytes)